

Indicators for detection of abusive digital lenders.

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Context

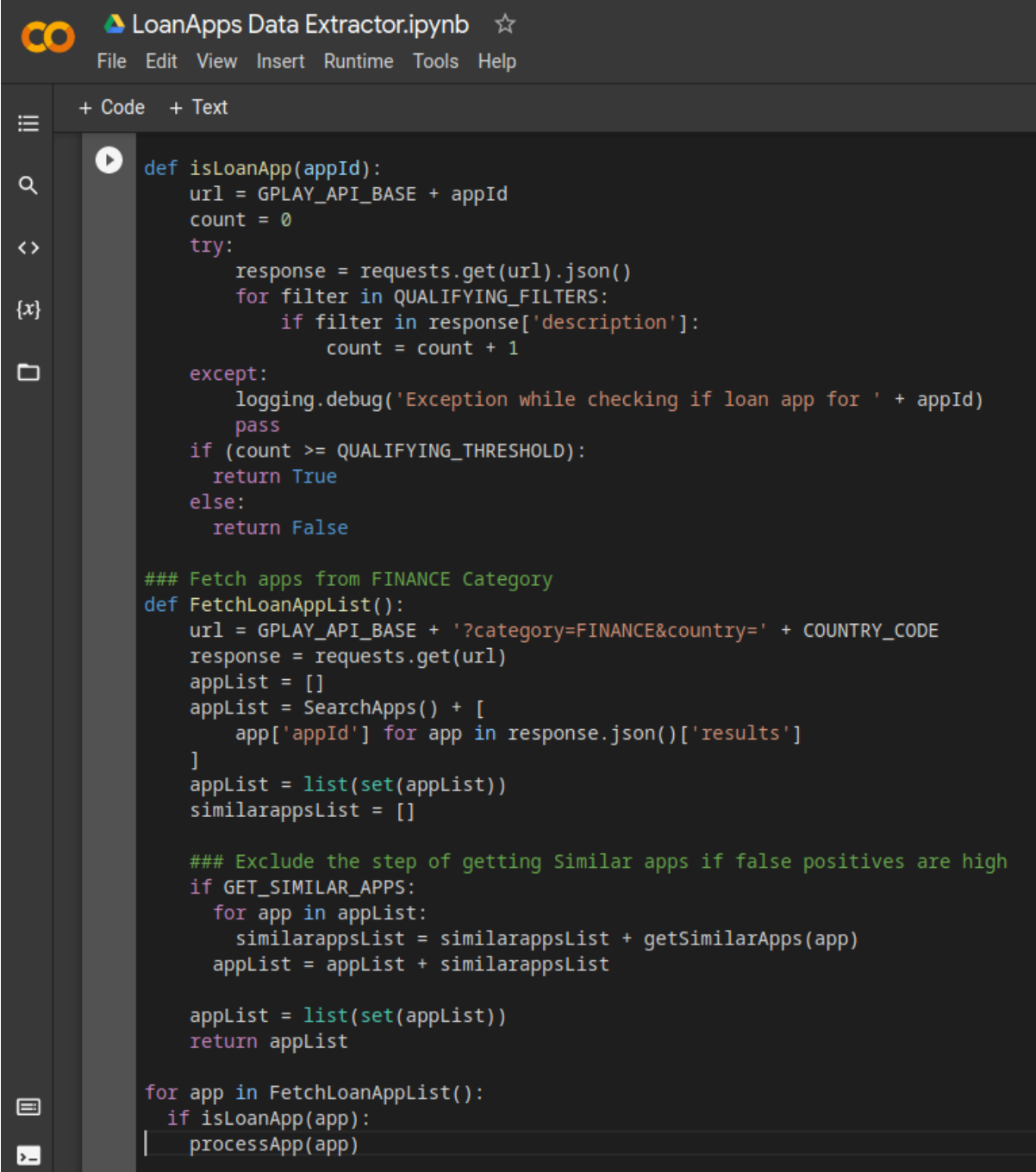
Digital lending is growing rapidly and along with the growth there has been a noticeable rise in consumer harms. These harms arise due to a variety of factors such as regulatory void on upcoming technology and data use in the financial sector, digital literacy, financial literacy leading to exploitative practices and suboptimal outcomes such as eroding consumer trust. Identification of abusive practices to overcome the harms hence becomes a key issue towards a common good for everyone in the ecosystem. This brief aims at documenting a list of indicators to spot abusive digital lenders from publicly available metadata around digital loan apps.

Data sources available

1. Google Play - Source metadata about apps by country, metadata about each app, their permissions, reviews, etc.
2. List of 3rd party providers offering services to digital lenders.

Methodology

1. Fetch list of loan apps from Google Play FINANCE category in the country (e.g. Nigeria), search results in Google Play store for popular search terms and similar apps. This could also be used to run for any country and get a list of loan apps.



```
def isLoanApp(appId):
    url = GPLAY_API_BASE + appId
    count = 0
    try:
        response = requests.get(url).json()
        for filter in QUALIFYING_FILTERS:
            if filter in response['description']:
                count = count + 1
    except:
        logging.debug('Exception while checking if loan app for ' + appId)
        pass
    if (count >= QUALIFYING_THRESHOLD):
        return True
    else:
        return False

### Fetch apps from FINANCE Category
def FetchLoanAppList():
    url = GPLAY_API_BASE + '?category=FINANCE&country=' + COUNTRY_CODE
    response = requests.get(url)
    appList = []
    appList = SearchApps() + [
        app['appId'] for app in response.json()['results']
    ]
    appList = list(set(appList))
    similarappsList = []

    ### Exclude the step of getting Similar apps if false positives are high
    if GET_SIMILAR_APPS:
        for app in appList:
            similarappsList = similarappsList + getSimilarApps(app)
            appList = appList + similarappsList

    appList = list(set(appList))
    return appList

for app in FetchLoanAppList():
    if isLoanApp(app):
        processApp(app)
```

Screenshot of script

2. Fetch app metadata including the permissions data for all apps from the above list.
 - a. Following are the various attributes available for a given app as metadata, along with some notes

Attribute	Text	Notes
ID	Unique ID for the record	
Title	Title of the app	
Description	Description	
DescriptionHTML	Description in HTML	

Summary	Summary of the app	
installs	Installs as displayed on Google Play	Install data is gameable with bot downloads for higher listing in search, doesn't represent actual user base as users may have installed, but never used / installed via other app stores / sideloaded.
minInstalls	Minimum Installs	
maxInstalls	Approx * Maximum Installs	
score	Weighted average of ratings	
scoreText	Rounded off score displayed to user	
ratings	Total number of users who rated	
1 *	1 Star	Ratings and reviews are gameable too, usually are gamed widely across apps.
2 *	2 Star	
3 *	3 Star	
4 *	4 Star	
5 *	5 Star	
price	For paid apps	
IsFree	Free/Paid filter	0 for non-free, 1 for Free. Most apps are usually free.
currency	Currency for app pricing	
FreeText		
In App Purchase	In app purchase - with GPlay payment support options	0 for no IAP, 1 for IAP Supported. Most apps don't use IAP
Size	App Size	
androidVersion	Minimum Android API level	
androidVersionText	Textual android version support note	
Developer	Name of Developer	
Developer ID	Developer ID	
Developer Email	Developer Email	
developerWebsite	Developer Website	
developerInternalID	Internal ID of developer in Google Play	

Open Project Save Project Attach Database Close Database

Database Structure Browse Data Edit Pragmas Execute SQL

Table: loanapp_permissions Filter in any column

	id	permission	type	appId
	Filter	Filter	Filter	Filter
1	4827ff798763ba1176bb2fbfd6fba96c8f63f71e	read your contacts	Contacts	nga.docredit
2	13eca26bec88c8e22a005d8302ce67d225c5eb93	approximate location (network-based)	Location	nga.docredit
3	d8bbd4642f55bdac47ea29da883f9403c7a43884	precise location (GPS and network-based)	Location	nga.docredit
4	9f7450b2093c8e9837cf7b2ce81d4bfe150342c	read your text messages (SMS or MMS)	SMS	nga.docredit
5	b6c2e43b45fe419d50873e502be2cbb9719562...	directly call phone numbers	Phone	nga.docredit
6	aab34604f75c13d5fa32e71d29fb34b1e46a1be3	read phone status and identity	Phone	nga.docredit
7	cde68dacea129c3b86f9a0ea46f0c5fb83ce4da0	read the contents of your USB storage	Photos/Media/Files	nga.docredit
8	661918397bfcfaeacdced5d46083c8fca05f92fa9	modify or delete the contents of your USB ...	Photos/Media/Files	nga.docredit
9	72f719b88e9db1077c5abac98a2d1bf29a49f693	read the contents of your USB storage	Storage	nga.docredit
10	995b85a546c5f44c496c083e28cd510e0b86c718	modify or delete the contents of your USB ...	Storage	nga.docredit
11	6d7592ee870a8d58541b7a313ce6a00f51140c...	take pictures and videos	Camera	nga.docredit
12	fcf4e76dc906b541f91079cca879e7bab6f7ea9e	view Wi-Fi connections	Wi-Fi connection information	nga.docredit
13	4b96727c1a838986bbc0fc9e91e45995b61158f6	read phone status and identity	Device ID & call information	nga.docredit
14	cc9b9c33f3ebdad9c8ab5090bf6a5d587c6bbdb2	view network connections	Other	nga.docredit
15	07a244fa492ec2e89f37f527b2570edd1645d429	pair with Bluetooth devices	Other	nga.docredit
16	a3da7d2a7f821cc806764530d119409eb835c727	full network access	Other	nga.docredit
17	f24a38a904b5ac683599fd15a0c9d816891349e1	take pictures and videos	Camera	com.niri.nextcredit
18	05d79dbc0a33adbfc3dfacd8ca0056744dffccc6	approximate location (network-based)	Location	com.niri.nextcredit
19	44ec1929bc122e289b23033057659d09fdab80...	precise location (GPS and network-based)	Location	com.niri.nextcredit
20	d677b80e1741c1edfa24ef8c916af2ee486596fb	read your text messages (SMS or MMS)	SMS	com.niri.nextcredit
21	ed06e7ea121b13b615106f84cae46822732ad...	read phone status and identity	Device ID & call information	com.niri.nextcredit
22	7968b28737817172b1d0032e1b7cf9e6383c72...	read the contents of your USB storage	Storage	com.niri.nextcredit
23	1778c846aa2d0ddafde997e110a4b41aa61712ef	modify or delete the contents of your USB ...	Storage	com.niri.nextcredit

Screenshot of permissions table of the database

3. Additionally perform **malware analysis** on the app to list the communicating servers to build a 3rd party provider database of loan apps.
 - a. Single app takes 15-20 minutes to manually go through the data (scope for automating this process but this would then still need to be analysed).
 - b. Indicative list of third parties in India, identified after malware analysis of loan apps. Constant monitoring of apps to identify vendor supply chain is key to mapping the support ecosystem and spot bad actors early on.

	QueryName	QueryText
	Filter	Filter
1	Epoch	url:'sdk.epoch-api.com'
2	MoneyBox	url:'9443V/downloadlink'
3	AdvanceAI	url:'10.1.10.31\9825'
4	MoneyBox	url:'9443/downloadlink'
5	Hyperverge PAN	url:'ind-docs.hyperverge.co/v2.0/readPAN'
6	Hyperverge Aadhaar	url:'ind-docs.hyperverge.co/v2.0/readAadhaar'
7	Hyperverge FaceID	url:'faceid.hyperverge.co'
8	Kazra	url:'app.karza.in'
9	JuicyScore	url:'api.juicyscore.com/getscore'
10	Epoch-1	url:'sdk.epoch-api.comVapi'
11	CrediTech Aadhaar	url:'ai.creditech.bizVaadhaar_img_aV'
12	Algo360	url:'iapi.algo360.com'
13	ZipNACH	url:'uat1.zipnach.com'
14	CreditVidya	url:'api.creditvidya.com'
15	Digio	url:'ext.digio.in'
16	Digio Mandate	url:'ext.digio.in/#/api-mandate-response'
17	Simpl	url:'approvals-api.getsimpl.com/my-ip'
18	CDAC ESign	url:'esignservice.cdac'
19	Accuauth	url:'cloudapi.accuauth.comVfaceV...
20	HC Android Bundler	url:'web/protocol_borrow.html'
21	H5 Android Bundler	url:'url:'h5/bankAccount.html'
22	Shanshad	url:'testapp.shansd.com'
23	Tongdun	url:'idfp.tongdun.net'

- Analyse the above data for indicators to spot potentially harmful lenders.

Overview of Indicators

1. **Brand, Advertising** – Trademark violation, phishing. *Automatable, with some manual review*

- From a consumer standpoint, the first interaction with a lender is through advertising and brand awareness at the app store / mobile ad / offline ad about the lending app. While disproportionate spend on marketing is a flag for the consumer, a trustable brand differentiates itself by providing the commitment to customer, while most apps that try to gain trust by piggybacking themselves with other established brands by similar sounding names, trademarks are an early red flag indicator.
- E.g:- CashTM (PayTM font, colour scheme, maybe trademark highjack)
- **Excessive App Store Optimisation (ASO)** - CashBean¹ (IN) – com.loan.cash.credit.easy.dhan.quick.udhaar.lend.game.jaldi.paisa.borrow.rupee.play.kredit
 - A similar ASO strategy being practised in Nigeria by KashCredit² - indicates re-use of same playbook / possibly same actors.

Actors are using the same playbook, but it may also be the same actors.

2. **App Metadata**

- **Release date** - Analysis of metadata from Google Play / other app stores could indicate if the app predates the legal entity which owns the app. In the real world, most apps get built after a legal entity is formed and exceptions to this rule are a high risk indicator that the app is run by anonymous / unidentified, potentially rogue actors. This could be automated by extracting and comparing the release date from Google Play and an incorporation date from a third party database.
- **Developer email ID** - This also indicates if the same entity owns the app / if the app is vendor managed / other partnership structures. Due diligence on entity data from these can help consumers spot rogue lenders.
 - *It may be possible to scrape companies data, to find out who is behind the apps, but it varies country to country. Also company registration process/scrutiny/process different across countries.*
- **Privacy policy** - The language of privacy policy and terms and conditions can also be used as indicators. While more often than not, these are legalese boilerplate texts, the rogue lending apps, don't host them on their domains and use Google docs / other static web hosting to host these pages.
 - Non-domain hosted privacy policies e.g. gDocs, are more suspect.
- **Public presence** - Website of the apps, their **professionalism** (Sites have unlinked / missing pages) is also a good proxy indicator if a loan app is run by a trustable entity / rogue lender.
 - Many sites have standard templates, and sometimes don't change the default settings, e.g. default privacy policy.

¹ [RBI cancels Certificate of Registration of PC Financial Services | Economy News | Zee News](#) - CashBean's parent entity was barred from digital lending after consumer conduct / money laundering incidents.

²

<https://play.google.com/store/apps/details?id=com.kashkash.fastmoney.lending.easypay.branch.instant.creditcard.quickonline.loanapp.nigeria>

3. Physical Presence

- **Address** - The address information in the app metadata can be used to identify if the entity has a physical presence in the country, supports communication with users in local language, complies with regulations, has a legally responsible point of contact for grievance handling. While some jurisdictions do allow partnership structures without need for physical presence, increasingly physical presence is being mandated for overseas apps.

4. Cyber Hygiene

- **Cyber Hygiene** broadly refers to the tech practices by the provider and this could indicate if consumers are protected in the digital world. **Frequency of app updates** can be a useful proxy indicator for product increments / iterative market feedback and a stale app (any more older than 90 days) is a red flag. This could indicate an outsourced or lack of active technical team, increasing susceptibility to platform breakdowns or cyber threats.
- **Infrastructure upgrades & responsiveness** to cyber incidents are also useful indicators of having made right investments into technology operations and that assures consumers of dealing with an entity that is respectful / concerned about user-data. These could be assessed using threat-intel repositories like VirusTotal.com etc

5. 3rd Parties

- **3rd parties** - A digital lender might outsource one / more / all functions to a range of 3rd parties that include tech intermediaries like KYC providers, data providers, payment processors and operational intermediaries like sales / recovery operations.
 - With additional time and manual investigation, areas to explore include:
 - Which 3rd parties are the go-to for predatory apps?
 - What is the cyber hygiene of 3rd parties?
 - Add list of 3rd party providers for India. The following is a list of 3rd parties identified in loan apps in India that were in use in 2021.

Steps	Service	Providers
Customer acquisition	Targeted ads via different channels	Google, Facebook, Youtube, Rummy sites, Ad SDKs
KYC and Fraud check	Checks for ID verification and facial recognition	IDfy (India) , Hyperverge (US) , Advance.ai (China) , DeepFinch (China) , Veri5Digital (India) , Link Face (China)
Risk scoring	Capture mobile and personal data to score borrowers	Credit Vidya (Indian), Tongdun (China), Flashcloud (China), Epoch (China), Risk Cloud (China) Juicy Score (Russia), Fintopia (China), Zhenrongbao CN

Loan disbursement	Plug into banking APIs to disburse money into borrowers' account	Finflux (Indian)
Loan management	Help manage the loan lifecycle	Finflux (Indian), Perfios (Indian)
Loan servicing	Integrate with SMS providers to remind about due dates	Kaleyra (US), Twilio (American),
Loan collection	Integrate with payment gateways to collect money from borrowers	Razorpay (Indian), Cashfree (Indian)
Digital Loan collection	Collection bots that make automated calls to overdue borrowers	Flashcloud (China),

- Analysis of data practices and cyber hygiene of 3rd parties is a useful indicator of the potential to cause unintended consumer harm.
- **Whitelabel providers** - Analysis of apps can also indicate use of **whitelabel providers** and tracking history of whitelabel providers can also indicate structural harms that have happened in the past and how it was handled.

6. App Features (*mostly manual, with some automation*)

- **App permissions** give a broad overview of the capabilities the lender is interested in and mapping of these across ecosystems can indicate if a lender is too pesky on data / wants more control on the borrower.
 - *A strict*
 - In Nigeria, apps used 16 permissions on average, but some lenders had up to 40 permissions. A script could be written to extract and summarise key statistics.
 - *Summary if possible*
- **Clear UX** - Consent on data use is non-negotiable for a responsible lender. Additionally, clear terms of repayment should be disclosed upfront.
- **Document trail** - Email / SMS - multi-channel documentation delivery.
- **Background activity** - Apps with high background activity may be constantly pushing data from the phone to the app's servers. This should require explicit consumer consent that is comprehensible, inclusive of data harms of such collected data.

7. Responsiveness

- **CRM** - Whether apps are actively responding to customer reviews or complaints can be codified on lender responsiveness. This could be automated via review of Google Play data.
- **Social media** - Similarly, one could analyse checking an app's responsiveness via social media channels. While this currently requires manual review, this could be automated via social media / sentiment analysis.

Deep dive of select indicators

1. Permission Analysis

1. While most apps use a common set of permissions for various reasons ranging from KYC to credit assessment to liveness detection, there have been apps that use excessive permissions either for data grab / constant post-loan surveillance / control of the device.
2. Derive insights on least used permissions, most used permissions and map it against stated / unstated use of such permissions to analyse the possibility of harm.

Example :-

1. Of the 100+ apps in Nigeria identified as lending app in Nigeria, following is a list of popular permissions used by the app.

read the contents of your USB storage	116
modify or delete the contents of your USB storage	116
read phone status and identity	116
view network connections	116
full network access	116
prevent device from sleeping	103
view Wi-Fi connections	96
take pictures and videos	95
precise location (GPS and network-based)	86
read your contacts	85
approximate location (network-based)	84
find accounts on the device	80
control vibration	60
connect and disconnect from Wi-Fi	57
read your text messages (SMS or MMS)	53

run at startup	47
read calendar events plus confidential information	34
add or modify calendar events and send email to guests without owners' knowledge	24
pair with Bluetooth devices	23
modify your contacts	21
modify system settings	21
record audio	17
change network connectivity	17
read your own contact card	16
receive text messages (SMS)	15
draw over other apps	15
directly call phone numbers	15
control flashlight	15
access extra location provider commands	15
retrieve running apps	12
read Google service configuration	12
access USB storage filesystem	12
access Bluetooth settings	11
read battery statistics	8
read your Web bookmarks and history	7
reorder running apps	6
read sensitive log data	6
read call log	5
control Near Field Communication	5

change your audio settings	5
set an alarm	4
send sticky broadcast	4
create accounts and set passwords	4
allow Wi-Fi Multicast reception	4
use accounts on the device	3
read sync settings	3
measure app storage space	2
change system display settings	2
add or remove accounts	2
Google Play license check	2
toggle sync on and off	1
read terms you added to the dictionary	1
make/receive SIP calls	1
disable your screen lock	1
close other apps	1
add words to user-defined dictionary	1

- **read your Web bookmarks and history** for instance is an excessive permission, but is being used by at least 7 apps. But this was cited in an [IMF paper](#) as a novel way in using new types of information for credit assessment.
- **add or modify calendar events and send email to guests without owners' knowledge** - Apps doing *anything* without the knowledge of the owner of the device, especially modifying calendar events is cause for concern. Specific usage of this permission needs to be studied in the local context.

2. 3rd Party Provider analysis.

1. A digital lending app integrates multiple 3rd parties which a lender might prefer to outsource the operations. It is also possible, for whitelabel provider - who manages a service offering such as "Lending App as a service" to any investor wanting to be present in the market.
2. The behaviour and customer experience of a lending app is also a function of customer experience of these individual outsourced components that is part of the offering.
3. The analysis of third party providers can give useful indicators on size of market outside of Google Play, number of apps and operators who have numerous apps based on the same template of 3rd party providers.
4. Analysing data flows can also determine the data-power differentials in the ecosystem and if 3rd parties can become powerful rent-seeking intermediaries and in some cases, actually be in the digital lending industry by publicly posing as 3rd parties to avoid liabilities.
 - a. [Advance.AI](#) is a global facial recognition based KYC provider operating in multiple markets and has been building a private facial recognition based credit score product. Given the governance void around facial recognition and data protection in many of the markets they operate, a powerful data intermediary poses risk not just to consumers, but also to competition.
 - b. Geo-politically, a large number of 3rd parties in the space operate out taxation-lite / neutral locations such as Singapore / Hongkong, but have strong ties / base in China. Use of 3rd parties will increasingly be subject to geo-political scrutiny in a data-localisation based regime preferred by certain governments as geo-political strategy.
5. The re-use of standard tech-stack across multiple geographies also pose the risk of dumping as tech that is outlawed / out-regulated in certain geographies will be dumped in markets without similar protection. Prevalence of such dumpware poses clear threats to consumers as they use tech that has clear issues that have been witnessed elsewhere is still put in use.
 - a. Example - [Tongdun](#) is a Chinese fintech company that operated in China, moved to Indonesia when faced with challenges in Chinese regulation and have since moved to India, again when faced with regulatory challenges in predatory practices. The re-use of the tech infra is clearly spottable from Indian loan apps that communicate to <https://idfp.tongdun.net/> - a risk scoring instance setup for Indonesia originally.

3. App Metadata Analysis.

The app metadata obtained from playstore could be subject to analysis to derive various insights. Some examples are included below of useful inferences where data could show light and lead investigators to spot potentially harmful apps.

1. A view of privacy policy URLs and developer Website URLs can be useful indicators. Privacy policy pages hosted on Google docs / sites / other static web hosting services imply the digital lender doesn't have a website of their own, is most likely an informal player, by extension could be outside formal regulatory supervision / industry self regulation. The privacy policy URLs could also reveal if the lender is using a white label, their origin. <https://apph5.ncreditapp.com/yinsizhengce.html> indicates a Chinese whitelabel provider behind the app.

appid	Title	Privacy Policy
com.nigeria.aimloan	Aimloan-Reliable Loans Online	https://www.aimloan.ng/privacy
com.beatha.mkopokaka	Umba Mobile - Leading Digital Bank for Africa	https://www.umba.com/transparency?tab=tab-policy
uni.UNI4E5FF20	Nigcash	http://treaty.nigcash.com/api/info/privacy
com.whzkkjgs.procashme	Procash-instant loan online	https://www.procashme.com/agreement/appPrivacyPolicy.html
com.ng.ncash	NCash-Personal & Flexible Loan	https://app-api.9ncash.tech/wap/agreement?name=register&system=ncash
com.instant.cash.loan.payday.money.nigeria.new.c	CashCredit	https://www.cashcreditapp.com/article/privacy
com.naira.hub.financial.payment	Naira Hub	http://admin.hubnaira.com/NairaHub.html
com.quick.naira.cash.loan.app.opay.ngn.palmpay.c	Personal Loan app-BitCoin Loan	https://static.houseloansng.com/pages/common/ng15/ng15_policy/index.html
com.nigeria.addmoney	Addmoney-Reliable Loans Online	https://www.Addmoney.ng/privacy
com.niri.nextcredit	NextCredit - Fast loan, online cash in nigeria	https://sites.google.com/view/nextcredit-privacy
com.imoney.credit.nira	iMoney	https://imoney.ng/?page_id=208
com.oasis.ball	FoundOne - online loan app in nigeria	https://docs.google.com/document/d/1xRZ1v6Ge9OUvGz00cJD0TWQxMj1Dfstl
com.firstbank.lit	LIT by FirstBank	https://www.firstbanknigeria.com/home/legal/privacy-policy/
com.gold.nairavenue	Nairavenue-Instant loan	https://api.nairavenue.com
com.kredibank	Kredi Bank - Savings and Loans	http://www.thekredibank.com/privacy_policies.html
cc.flashnaira.app	FlashNaira	https://www.flashnaira.cc/privacy
com.loancash.android	Lucky loan	http://help.tekmkt.com/luckyloan/privacy.html
com.needsalaryinfonow.com	Need Salary Instant Loan App	https://pages.flycricket.io/need-salary-instant/privacy.html
com.niridaichao.moneyhub	MoneyHub - Loan App	https://api.ngmoneyhub.com/H5/html/termsCondition.html

- Age of the app, number of versions can indicate if the app gets sufficient updates necessary to keep it in sync with customer / market needs, technically safe and sound. Any app that has not received an update in the last 3 months indicates the entity behind is vulnerable to cyber risks.

3rd Party Digital infrastructures for lending

There are third parties who provide provider-level infrastructure, and ecosystem-level infrastructure. Besides market led software stack for various components of digital lending like KYC, alternate data, payments, increasingly a number of bank-led / ecosystem-led digital infrastructures are being built to enable digital credit, actively furthered by regulators / government. These technology infrastructures are designed to further machine-first, data-driven lending for various use cases, often originally aimed at addressing data-access / level playing field in the market for non-banks in the lending landscape.

While new technological infrastructure holds high potential for expanding financial access and improving lending decisions, they also pose potential harms which must be considered and addressed at their outset. Besides the generic harms arising out of algorithmic lending / fairness in AI/ML systems and data-driven credit decisioning, use of these infrastructures also enable adding a whole range of data-intermediaries into the lending value chain and brings its own risk.

S.No	Infrastru cture	Function	Opportunity	Risk / Consumer harm
1.	Account Aggregat or ³ (AA)	Consented financial data sharing	Access to trustable, systemic financial data.	Heightened access to systemic banking information, leading to higher demand for

³ <https://datagovernance.org/files/research/1641969121.pdf> Tech Tools to Facilitate and Manage Consent: Panacea or Predicament? A Feminist Perspective - Jain Tripti - Data Governance Network.

				data / over-consenting for credit access.
2.	Open Credit Enablement Network (OCEN) ⁴	Orchestrated lending via Loan Service Providers	Further embedded lending.	Newer forms of data being tapped for credit access and a large number of intermediaries.
3	* <i>Public Credit Registry</i> ⁵	Credit Registry of all debt	Attempts to solve for fixing thin-file credit history regulatorily, at population scale.	Lack of clear individual insolvency regime / clean-start norms leading to exclusion.

LoanApps Database Generator

The process of building the database of loan apps, getting a dump of metadata could be automated using [Google Play API](#) and data analysis on the metadata could be used to generate insights such as those mentioned above.

In addition, the list of apps could also be subjected to semi-automated malware analysis and additional metadata related to 3rd parties could be generated and subjected to analysis, deriving insights.

⁴ <https://github.com/iSPIRT/OCEN> Loan Service Provider (LSP) <-> Lender protocol specification

⁵ [RBI: RBI likely to launch public credit registry next year, create a hub of borrowers' info, BFSI News, ET BFSI](#) - Public Credit Registry by the Indian banking regulator is expected to roll out in 2023-24

LoanApps Data Extractor.ipynb

File Edit View Insert Runtime Tools Help All changes saved

Files

..

sample_data

loanapps.db

loanappsdata_NG.db

[1] !pip install flatten_json sqlite_utils

Initialize Country Code, Searchterms and Qualifying Filters, Crawler options

COUNTRY_CODE = 'NG'

Popular search terms used by users in Google Play store

SEARCH_TERMS = [

'loan', 'personal loan', 'salary loan', 'loan naira', 'e-loan',

'borrow money', 'Sokoloan', 'Lapo'

]

List of words that act as a filter to restrict lending apps of a particular

geography by setting specific keywords to lookout in the app description

QUALIFYING_FILTERS = ['loan', 'Loan', 'credit']

Minimum Number of qualifying filters to pass the

threshold to consider an app is a digital lending app

QUALIFYING_THRESHOLD = 1

Specify if the crawler must look for similar apps for each app.

Takes longer, risk of higher false positives. Options True/False

GET_SIMILAR_APPS = False

DB_FILENAME = 'loanappsdata_' + COUNTRY_CODE + '.db'

Database Generator

Generate loan app database from searching FINANCE category apps in a specified country with listed search terms meeting qualifying word filters in their description above a threshold.

[] 2 cells hidden